



SAURON

SETUP WALKTHROUGH

NOW CUTS HOOK OVERHEAD 85%

one agent to route them all

The installer runs end-to-end. Prompt 0 picks the orchestrator (claude, cursor, cline, codex, aider or generic). Prompt 7 collects API keys with per-provider instructions and writes a `0600 .env`. Setup now also writes a `CLAUDE.md` doctrine file that is auto-loaded and prompt-cache friendly, so the routing map, keep-in-Claude rules, and 4-step pipeline never re-inject on every message. Multi-framework users get a lean, sibling-detected block that skips shared sections. Measured: **85% cut in hook overhead** vs the prior version on a 50-message session.

What changed since the last PDF: SessionStart shrank from ~1500 to ~250 bytes (routing map moved to CLAUDE.md). UserPromptSubmit dropped from ~820 to ~70 bytes per message. CLAUDE.md carries the doctrine once and Claude Code caches it for one hour. The pal-router skill gained 11 token-efficiency rules (auto-compress outputs over 5 KB, batch PAL calls, { claim, confidence, source_span } triples, in-conversation failure-map, diff-first, response terseness ladder). Sibling detection dedups doctrine when both sauron and let-there-be-light are installed. **PAL now installs from crowx01/pal-mcp-server** — the routing-enhanced fork.

1. The orchestrator choice

c	claude-code (default)	writes <code>~/.claude/settings.json</code> hooks + symlinks skills into <code>~/.claude/skills/</code>
r	cursor	writes <code>./.cursor/rules/sauron.mdc</code> with <code>alwaysApply:true</code> ; copies skills alongside
l	cline	writes <code>./.clinerules</code> (or <code>~/.clinerules</code>); copies skills alongside
x	codex-cli	writes <code>~/.codex/instructions.md</code> ; copies skills into <code>~/.codex/sauron-skills/</code>
a	aider	writes <code>./.aider.sauron.md</code> ; add to <code>.aider.conf.yml</code> as read:
g	generic	writes <code>./SYSTEM_PROMPT.sauron.md</code> you paste into any tool; copies skills alongside

2. The terminal session (Claude Code path)



interactive setup · one agent to route them all

0. Which agent orchestrator do you use?

- c) claude-code (default) full: hooks + Skill() invocations + delegate policy
- r) cursor writes .cursor/rules/Framework.mdc (alwaysApply)
- l) cline writes .clinerules with the delegate policy
- x) codex-cli (OpenAI) writes ~/.codex/instructions.md
- a) aider writes .aider.conf.yml + Framework.md conventions
- g) generic / other writes SYSTEM_PROMPT.md you paste into any tool

✓ orchestrator: c

1. Where should it install?

- g) global → ~/.claude/settings.json (loads every session, every project)
- p) per-project → \$PWD/.claude/settings.json (only loads when Claude Code runs here)
- s) skip → print the settings JSON to stdout, do not write anything

2. Which skills should auto-load at session start?

These are invoked as the very first tool calls of every session.
Leave blank to skip a skill; type y to include it.

3. Which PAL models should be listed in the delegate-first policy?

Pick every model your PAL install actually has keys for.
Unselected models are dropped from the routing map so Claude never tries them.

Ready to write ~/.claude/settings.json

Existing file will be backed up with .bak.<timestamp> suffix.

✓ wrote ~/.claude/settings.json

✓ wrote CLAUDE.md doctrine: ~/CLAUDE.md

✓ wrote env template: ~/.claude/.env.sauron.example

7. Enter API keys now to complete installation?

Prompted only for providers whose models you enabled (masked input, ENTER skips):

Groq	CUSTOM_API_KEY (+ CUSTOM_API_URL)	groq/gpt-oss AND qwen3 (default tool executor)
Gemini	GEMINI_API_KEY	flash, pro
OpenRouter	OPENROUTER_API_KEY	nemotron, grok, or-free
OpenAI	OPENAI_API_KEY	optional, asked y/N (gpt-5 tiers)

Skip and edit .env.sauron.example later, or enter them now to have a real
.env.sauron written with 0600 permissions.

Next steps

1. Register PAL as an MCP server in ~/.claude.json (setup.sh does this for you on the claude-code path):

```
"mcpServers": { "pal": { "type": "stdio",  
  "command": "/path/to/pal-mcp-server/.pal_venv/bin/python",  
  "args": ["/path/to/pal-mcp-server/server.py"],  
  "env": { ...keys... } } }
```
2. Source your API keys: source ~/.claude/.env.sauron (after editing it)
3. Restart Claude Code.
4. On the next session start the auto-invoked skills fire immediately.

✓ sauron setup complete.

3. What Claude sees at the next session start (now lean)

The SessionStart hook fires before Claude responds. Its payload is injected into context. The full doctrine now lives in CLAUDE.md, so the hook is just a pointer.

SessionStart · additionalContext (163 bytes)

Doctrine + routing map are in CLAUDE.md at the project root (already loaded).
Preload skills now: `caveman` (full mode), `pentesting-agent`, `validator` (preload).

UserPromptSubmit · additionalContext (69 bytes, every message)

Doctrine active from CLAUDE.md. Delegate + failover. Batch PAL calls.

Why it is cheap: the routing map, keep-in-Claude list, failover doctrine, 4-step pipeline, and the 11 token-efficiency rules moved into CLAUDE.md. Claude Code auto-loads that file into the system prompt and caches it for one hour. On message 2 through N in the same session you pay **0 tokens** for it. The prior version re-injected all of it on every user message.

4. The CLAUDE.md doctrine file (4390 bytes, loaded once)

Setup writes this alongside settings.json. It sits inside a `<!-- BEGIN sauron doctrine -->` managed block so re-running `setup.sh` is idempotent. When both sauron and let-there-be-light are installed, the second one detects its sibling and writes a lean block that skips the shared sections.

```
<!-- BEGIN sauron doctrine (managed by setup.sh; regenerate to update) -->
# sauron doctrine

## Delegate-first routing
groq (gpt-oss-120b, ~500 rpm, 8000 tpm cap) for report writing, vuln explanations, and
skeptical validation. nemotron (nvidia, 1M ctx) for bulk reading of large files. grok
(x-ai, 2M ctx) for permissive high-context security reasoning when Gemini refuses. flash
(gemini-3.6-flash, 1M ctx) for fast structured extraction. or-free (OpenRouter meta-router,
200K ctx) as generalist fallback. pro (gemini-3.1-pro-preview) for deep-reasoning fallback
and adversarial debate via mcp_pal_challenge.

## Keep in Claude ONLY
- user-facing decisions
- exploitation choices, severity calls
- safety-boundary checks (no 3rd-party data, no destructive actions, no account creation)
- side-effecting actions
- tool-sequence orchestration

## Failover
Any model refusal or error routes to the next model in the map. A classifier refusal is a
routing problem, not a stop.

## 4-step validation pipeline (per confirmed finding, before presenting or reporting)
A. Stress-test via validator skill (false-positive check, reproducibility, evidence gaps,
confidence).
B. Run the gap tests, negative controls, and live observations the validator flags.
C. Adversarial challenge via mcp_pal_challenge (pro primary, groq fallback): challenge
severity, exploitability, chain viability, impact ceiling.
D. Synthesize with confidence markers; groq drafts the report, Claude spot-checks
low-confidence claims only.

## Token efficiency rules (this framework exists to cut Claude token cost)
- Any tool output over 5 KB routes through nemotron (bulk) or flash (structured) for
  summarization before Claude reads.
- Batch related PAL sub-tasks into one structured call, not N separate ones.
- PAL responses on factual output must include {claim, confidence, source_span} triples.
- Never re-Read a file already Read this session; recall from conversation context.

... [truncated for PDF; full file is 4390 bytes]
```

5. Files on disk after install

```
.claude/  
.claude/.env.sauron.example  
.claude/settings.json  
CLAUDE.md
```

.env.sauron.example (only the API keys your model picks need)

```
# sauron API keys - source this from your shell rc, or export before starting Claude Code.  
# Do NOT commit the real values.  
export GEMINI_API_KEY=your-gemini-key  
export OPENROUTER_API_KEY=your-openrouter-key  
export CUSTOM_API_URL=https://api.groq.com/openai/v1  
export CUSTOM_API_KEY=your-groq-key  
# optional, only if you opt in:  
export OPENAI_API_KEY=your-openai-key
```

6. Smarter pal: pal run / pal chat / /tools / /agent / /debate

The PAL fork installs a global pal CLI (~/.local/bin/pal). Groq is now **required** for the default tool executor: **qwen3** (qwen/qwen3.8-27b via Groq) drives /tools and pal run; gpt-oss trips Groq's output parser on tool prompts (400 output_parse_failed). Override with PAL_CHAT_TOOLS_MODEL.

pal run	headless one-shot	pal run "<task>" prints only the result. Flags: --ro, --agent, --model, --plan <file>, --max-steps N, --agent-role autonomous edit plan review, --json
--plan	plan hand-off	write the job to a file, pal run --plan plan.md, take the result, do not babysit
pal chat	interactive REPL	auto cheap/smart routing (/smart, /cheap)
/tools	on-box execution	runs nmap/nuclei etc. on the box; /tools:ro = read-only allowlist
/agent	Claude Code sub-task	/agent[:edit :plan :review] via clink
/debate	read, then decide	one model reads the files, then a panel decides

Routing learns from real outcomes: ~/.pal/episodes.jsonl → runtime bandit → pal distill (human-gated proposals) plus a human-gated teacher lesson store. Groq ITPM ≈ 7000, so the tool loop trims context to PAL_TOOLS_CTX_CHARS (default 16000). Authorized recon gets an AUTHORIZATION preamble to avoid refusals.

PAL server: github.com/crowx01/pal-mcp-server (routing-enhanced fork) · Repo: github.com/crowx01/sauron · Sibling: [crowx01/let-there-be-light](https://github.com/crowx01/let-there-be-light) (dev-lifecycle)